CORE Element Catalog v1.0

Centrally Organized Relational Emergence — Full Toolbox Manifest

Every Element listed here implements the CORE Connection Standard. Orbs dock via Umbilical. Satellites tether via Tether. Formulas compose multiple Elements into a single deployable unit. External Connectors are Portal destinations reached through the CORE boundary.

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1. ORBS (20)

Orbs are the primary capability units. Each docks to a CORE Umbilical socket and carries a distinct, self-contained function.

Development & Code

#	ID	Name	Description	Primary Capability	Memory	Threads
1	orb-codegen	Code Generation Engine	Generates source code from intent descriptions, templates, and relational context. Supports multiple languages and frameworks.	`code_generation`	512 MB	4
2	orb-refactor	Refactoring Loom	Analyzes code structure and applies transformation patterns — rename, extract, inline, restructure — while preserving relational contracts.	`code_refactoring`	384 MB	2

#	ID	Name	Description	Primary Capability	Memory	Threads
3	orb-ast	Abstract Syntax Weaver	Parses source into AST, enables structural queries and transformations. The foundation other code Orbs build on.	`ast_parse`, `ast_transform`	256 MB	2
4	orb-debug	Fault Trace Navigator	Traces execution paths, identifies fault origins, maps error propagation through relational chains.	`fault_trace`, `stack_analysis`	384 MB	2
5	orb-testgen	Test Synthesis Engine	Generates test cases from contracts, interfaces, and invariant declarations. Produces unit, integration, and property-based tests.	`test_generation`, `invariant_test`	512 MB	4

Data & Knowledge

#	ID	Name	Description	Primary Capability	Memory	Threads
6	orb-schema	Schema Crystallizer	Defines, validates, and migrates data schemas.	`schema_define`, `schema_migrate`	256 MB	1

#	ID	Name	Description	Primary Capability	Memory	Threads
			Tracks schema evolution across versions without breaking contracts.			
7	orb-query	Relational Query Engine	Constructs and optimizes queries across structured and graph-based data sources.	`query_build`, `query_optimize`	384 MB	2
8	orb-index	Knowledge Indexer	Builds and maintains search indices over project artifacts — code, docs, contracts, telemetry logs.	`index_build`, `semantic_search`	512 MB	2
9	orb-transform	Data Transform Pipeline	Applies ETL-style transformations: map, filter, aggregate, join, reshape. Operates on streaming or batch data.	`data_transform`, `stream_process`	384 MB	4
10	orb-embed	Embedding Lattice	Generates vector embeddings for code, text, and structural elements.	`embedding_generate`, `similarity_score`	768 MB	4

#	ID	Name	Description	Primary Capability	Memory	Threads
			Feeds semantic search and similarity analysis.			

Infrastructure & Operations

#	ID	Name	Description	Primary Capability	Memory	Threads
11	orb-deploy	Deployment Conductor	Orchestrates build, package, and deployment pipelines. Manages environment configs and rollback states.	`deploy_orchestrate`, `rollback`	384 MB	2
12	orb-monitor	Telemetry Witness	Observes runtime behavior without interfering. Collects metrics, traces, and logs. Never modifies the system it watches.	`telemetry_collect`, `trace_observe`	256 MB	1
13	orb-env	Environment Crystallizer	Manages environment variables, secrets references, and configuration state across dev/staging/prod. Never stores secrets	`env_manage`, `config_resolve`	128 MB	1

#	ID	Name	Description	Primary Capability	Memory	Threads
			directly — holds references only.			
14	orb-container	Container Forge	Builds, manages, and orchestrates container images and runtimes. Dockerfile generation, compose orchestration, registry interaction.	`container_build`, `container_orchestrate`	512 MB	4

Communication & Integration

#	ID	Name	Description	Primary Capability	Memory	Threads
15	orb-api	API Boundary Engine	Generates, validates, and documents API surfaces — REST, GraphQL, gRPC. Enforces contract-first design.	`api_generate`, `api_validate`	384 MB	2
16	orb-webhook	Event Relay Conduit	Manages inbound and outbound webhook registrations, payload validation, retry logic, and dead-letter handling.	`webhook_manage`, `event_relay`	256 MB	2

#	ID	Name	Description	Primary Capability	Memory	Threads
17	orb-auth	Identity & Access Lattice	Handles authentication flows, token lifecycle, permission models, and role-based access patterns. Never stores credentials directly.	`auth_flow`, `permission_model`	384 MB	2

AI & Reasoning

#	ID	Name	Description	Primary Capability	Memory	Threads
18	orb-intent	Intent Synthesis Engine	Interprets natural language intent and maps it to CORE operations, Element selections, and connection plans.	`intent_parse`, `action_plan`	768 MB	4
19	orb-review	Code Review Arbiter	Performs multi-dimensional code review: correctness, security, style, performance. Returns findings, never verdicts.	`code_review`, `finding_report`	512 MB	4
20	orb-doc	Documentati	Generates	`doc_gener`	256 MB	2

#	ID	Name	Description	Primary Capability	Memory	Threads
		on Weaver	and maintains documentation from code, contracts, and architectural relations. Keeps docs in sync with reality.	`ate`, `doc_sync`		

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2. SATELLITES (30)

Satellites tether to Host Orbs and provide specialized sub-capabilities. Each belongs to one of five classes. Satellites can Mesh with other Satellites for shared services.

I/O Class (6)

#	ID	Name	Description	Mesh Services Offered
1	sat-stdin	Standard Input Receiver	Captures and buffers user input streams — keyboard, clipboard, file drops.	`input_stream`
2	sat-stdout	Standard Output Emitter	Formats and delivers output — terminal, UI panel, file write, notification.	`output_stream`
3	sat-fileio	File System Relay	Reads, writes, watches, and diffs files on the local filesystem. Handles encoding detection and line-ending normalization.	`file_read`, `file_write`, `file_watch`
4	sat-netio	Network I/O	HTTP/HTTPS	`http_request`,

#	ID	Name	Description	Mesh Services Offered
		Conduit	client, WebSocket connections, Server-Sent Events. Raw network transport layer.	`ws_connect`, `sse_listen`
5	sat-clipboard	Clipboard Bridge	Reads from and writes to the system clipboard. Handles text, rich text, images, and file references.	`clipboard_read`, `clipboard_write`
6	sat-terminal	Terminal Emitter	Spawns and manages shell processes, captures stdout/stderr, sends stdin.	`shell_spawn`, `shell_io`

Compute Class (6)

#	ID	Name	Description	Mesh Services Offered
7	sat-parse	Syntax Parser Satellite	Language-aware parsing: tokenizer, lexer, tree-sitter integration. Feeds AST data to host Orbs.	`token_stream`, `parse_tree`
8	sat-lint	Lint Evaluator	Runs linting rules against source, reports violations with fix suggestions. Configurable rule sets.	`lint_report`, `autofix_suggest`
9	sat-format	Code Formatter	Applies formatting rules — Prettier, Black, rustfmt, gofmt — without changing semantics.	`format_apply`
10	sat-compile	Compilation	Invokes language	`compile_invok`

#	ID	Name	Description	Mesh Services Offered
		Worker	compilers/transpilers. TypeScript, Rust, Go, C++. Returns diagnostics.	`e`, `diagnostic_stream`
11	sat-minify	Asset Minifier	Minifies JS, CSS, HTML, SVG. Tree-shakes dead code. Produces source maps.	`minify_apply`, `sourcemap_generate`
12	sat-bundle	Module Bundler	Resolves module graphs, bundles assets, handles code splitting and lazy loading.	`bundle_build`, `chunk_split`

Storage Class (6)

#	ID	Name	Description	Mesh Services Offered
13	sat-git	Git State Satellite	Reads git state: status, diff, log, blame, branches, stash. No destructive operations — read and stage only.	`git_status`, `git_diff`, `git_log`
14	sat-cache	Ephemeral Cache Layer	In-memory key-value cache with TTL. Shared across meshed Satellites.	`cache_get`, `cache_set`, `cache_invalidate`
15	sat-snapshot	State Snapshot Keeper	Captures and restores point-in-time snapshots of Element state. Enables undo/redo across operations.	`snapshot_save`, `snapshot_restore`
16	sat-log	Structured Log Sink	Receives, indexes, and queries structured log entries. Retains	`log_append`, `log_query`

#	ID	Name	Description	Mesh Services Offered
			within session only — nothing persists.	
17	sat-artifact	Build Artifact Store	Manages build outputs — binaries, bundles, images, test reports. Tracks provenance.	`artifact_store`, `artifact_retrieve`
18	sat-queue	Message Queue Satellite	FIFO and priority message queue for inter-Satellite async communication.	`queue_enqueue`, `queue_dequeue`, `queue_peek`

Validation Class (6)

#	ID	Name	Description	Mesh Services Offered
19	sat-typecheck	Type Checker Satellite	Runs type checking across TypeScript, Python (mypy), Rust, Go. Returns typed diagnostics.	`typecheck_run`, `type_diagnostic`
20	sat-test	Test Runner Satellite	Executes test suites — Jest, pytest, cargo test, go test. Streams results in real time.	`test_run`, `test_result_stream`
21	sat-schema-v	Schema Validator	Validates data against JSON Schema, OpenAPI specs, Protobuf definitions, database schemas.	`schema_validate`, `schema_diff`
22	sat-contract	Contract Verifier	Verifies that CORE Relation Contracts are correctly implemented and invariants hold at runtime.	`contract_verify`, `invariant_audit`

#	ID	Name	Description	Mesh Services Offered
23	sat-coverage	Coverage Analyst	Computes code coverage from test runs. Maps covered/uncovered paths. Identifies dead code.	`coverage_compute`, `dead_code_detect`
24	sat-regression	Regression Detector	Compares current test results against baseline. Flags regressions with bisect guidance.	`regression_detect`, `bisect_suggest`

Security Class (6)

#	ID	Name	Description	Mesh Services Offered
25	sat-scan	Vulnerability Scanner	Scans dependencies and code for known CVEs, OWASP patterns, and insecure configurations.	`vuln_scan`, `cve_report`
26	sat-secrets	Secret Leak Detector	Scans staged files, diffs, and outputs for accidentally exposed secrets — API keys, tokens, credentials.	`secret_scan`, `leak_alert`
27	sat-sandbox	Containment Boundary Guard	Enforces sandboxing constraints on child processes. Prevents privilege escalation.	`sandbox_enforce`, `permission_check`
28	sat-audit	Access Audit Trail	Records what accessed what, when, through which contract. Immutable within session.	`audit_record`, `audit_query`
29	sat-cors	CORS & Policy	Validates and	`cors_validate`

#	ID	Name	Description	Mesh Services Offered
		Enforcer	generates CORS headers, CSP policies, and HTTPS enforcement rules.	, `csp_generate`
30	sat-checksum	Integrity Verifier	Computes and validates checksums for contracts, artifacts, and data transfers. Detects tampering.	`checksum_compute`, `integrity_verify`

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3. FORMULAS (12)

Formulas are composed relational patterns that bind multiple Orbs and Satellites into a single deployable unit. A Formula is a recipe — it declares which Elements participate and how they connect.

#	ID	Name	Composition	Purpose
1	frm-fullstack	Full-Stack Scaffold	`orb-codegen` + `orb-api` + `orb-schema` + `sat-typecheck` + `sat-test`	Generates a complete full-stack application scaffold with typed API, schema, and tests.
2	frm-cicd	CI/CD Pipeline Formula	`orb-deploy` + `orb-container` + `sat-compile` + `sat-test` + `sat-artifact` + `sat-scan`	End-to-end continuous integration and deployment pipeline with security scanning.
3	frm-review	Comprehensive Review Formula	`orb-review` + `orb-ast` + `sat-lint` + `sat-typecheck` + `sat-coverage` + `sat-secrets` + `sat-scan`	Multi-dimensional code review: style, correctness, coverage, security, secrets.
4	frm-tdd	Test-Driven Dev Formula	`orb-testgen` + `orb-codegen` +	Write tests first, generate implementation to

#	ID	Name	Composition	Purpose
			`sat-test` + `sat-coverage` + `sat-regression`	satisfy them, verify coverage and regressions.
5	frm-api-first	API-First Design Formula	`orb-api` + `orb-schema` + `orb-doc` + `sat-schema-v` + `sat-typecheck`	Design the API surface and schema first, generate code and docs from the contract.
6	frm-debug	Deep Fault Analysis Formula	`orb-debug` + `orb-monitor` + `sat-log` + `sat-git` + `sat-snapshot` + `sat-regression`	Trace fault to origin, correlate with git history, compare against last known good state.
7	frm-migrate	Data Migration Formula	`orb-schema` + `orb-query` + `orb-transform` + `sat-schema-v` + `sat-snapshot` + `sat-checksum`	Safe schema migration with validation, rollback snapshots, and integrity verification.
8	frm-doc-sync	Living Documentation Formula	`orb-doc` + `orb-ast` + `orb-api` + `sat-git` + `sat-contract`	Generates documentation that stays in sync with code, API surfaces, and contracts automatically.
9	frm-secure	Security Hardening Formula	`orb-auth` + `orb-api` + `sat-scan` + `sat-secrets` + `sat-cors` + `sat-sandbox` + `sat-audit`	Full security pass: auth flows, vulnerability scan, secret detection, CORS, CSP, audit trail.
10	frm-refactor	Safe Refactoring Formula	`orb-refactor` + `orb-ast` + `sat-typecheck` + `sat-test` + `sat-coverage` + `sat-`	Refactor with full safety net: type checks, tests, coverage, and regression detection at every step.

#	ID	Name	Composition	Purpose
			`regression`	
11	frm-onboard	Project Onboarding Formula	`orb-index` + `orb-doc` + `orb-intent` + `sat-git` + `sat-log`	Indexes a new project, generates overview documentation, and enables intent-based navigation.
12	frm-embed-search	Semantic Search Formula	`orb-embed` + `orb-index` + `orb-query` + `sat-cache` + `sat-fileio`	Builds vector embeddings, indexes the codebase, enables natural language code search.

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4. EXTERNAL CONNECTORS (14)

External Connectors are Portal destinations. Each represents a real service outside the CORE boundary, reached through a Portal and its Tender conduit.

Portal connections enforce the same NegativeSpaceInvariants as any other contract. The connector handles protocol translation between the external service's API and the CORE's internal relational protocol.

#	ID	Name	Service	Capabilities Bridged	Auth Method
1	ext-github	GitHub Gateway	GitHub API v4 (GraphQL) + v3 (REST)	Repos, PRs, issues, actions, code search, reviews, releases	OAuth App / PAT
2	ext-gdocs	Google Docs Bridge	Google Docs API + Drive API	Document read/write, collaborative editing, export, folder management	OAuth 2.0
3	ext-claude	Claude Code Relay	Claude API (Messages, Tool Use, Agents)	Intent synthesis, code generation, review, multi-agent orchestration	API Key

#	ID	Name	Service	Capabilities Bridged	Auth Method
4	ext-slack	Slack Conduit	Slack Web API + Events API	Channel messaging, thread replies, file sharing, status updates, notifications	Bot Token
5	ext-linear	Linear Project Bridge	Linear GraphQL API	Issues, projects, cycles, labels, team sync, status tracking	API Key
6	ext-figma	Figma Design Relay	Figma REST API + Plugin API	Design read, component inspection, design tokens, code connect mappings	OAuth 2.0
7	ext-supabase	Supabase Data Gateway	Supabase REST + Realtime + Auth	Database CRUD, realtime subscriptions, auth management, storage, edge functions	API Key
8	ext-vercel	Vercel Deploy Conduit	Vercel API	Project deployment, environment management, domain config, build logs, serverless functions	Token
9	ext-npm	npm Registry Bridge	npm Registry API	Package search, publish, version management, dependency resolution, audit	Auth Token
10	ext-docker	Docker Hub Gateway	Docker Hub API + Registry API v2	Image push/pull, tag management, build triggers, vulnerability	Token

#	ID	Name	Service	Capabilities Bridged	Auth Method
				scanning	
11	ext-notion	Notion Knowledge Bridge	Notion API	Page/database read/write, block manipulation, search, template management	Integration Token
12	ext-gsheets	Google Sheets Relay	Google Sheets API	Spreadsheet read/write, formula evaluation, chart data, structured data import/export	OAuth 2.0
13	ext-sentry	Sentry Error Relay	Sentry API	Error tracking, issue management, release tracking, performance monitoring, alert config	DSN + Auth Token
14	ext-azure	Azure Cloud Gateway	Azure Resource Manager + various service APIs	Resource provisioning, deployment, monitoring, storage, compute, AI services	Service Principal

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5. ELEMENT COUNT SUMMARY

Category	Count	Connection Type
Orbs	20	Umbilical
Satellites	30	Tether + Mesh
Formulas	12	Composed
External Connectors	14	Portal
Total Elements	**76**	

The CORE has 8 Umbilical sockets. With Formulas composing multiple Elements, and Satellites tethering to docked Orbs, the full toolbox can operate through those 8 sockets. Mesh connections between Satellites create the emergent service fabric.